Research CLI — 系统架构 · Agent 请求数据流

Terminal UI → core orchestration & routing → tools / MCP / research modules → model providers

主数据流　　虚线框 = 架构分区

回退 / 反馈　　盒内 = 模块 + 职责

项目介绍

Research CLI 是终端优先的 AI Agent 工作流，将交互会话连接到模型、工作区工具和 MCP 服务。

核心路由负责上下文与模型供应商选择，研究模块覆盖论文写作、文献管理、数据分析和投稿准备。

CLI FRONTEND · packages/cli

- **Interactive TUI** — research.tsx · sessions
- **Commands** — mcp · skills · hooks
- **ACP** — acpSession.ts

CORE ORCHESTRATION · packages/core

- **Context & Memory** — config.ts · memory.ts
- **modelRouterService** — provider selection
- **Routing Strategy** — default · classifier · approval

TOOLS & INTEGRATIONS

- **Workspace Tools** — filesystem · shell
- **MCP** — external service adapters
- **Research Modules** — writing · bibliography · analysis
- **SDK / A2A / VS Code** — embedded clients

tool result loop

MODEL PROVIDERS · provider config

- **Gemini / Vertex** — hosted models
- **OpenAI / Anthropic** — hosted models
- **Baidu / DeepSeek / Qwen** — hosted models

RESULT STREAM

- **Tool Results** — structured observations
- **Assistant Response** — streamed to terminal

主链：prompt → packages/cli → packages/core router → provider / tools → structured result → terminal。